



BrightLearn Data Analytics

Research Assignment 1

Foundations of Data Analytics & Data Science

Course	Data Analytics & Data Science
Assignment No.	Research Assignment 1
Target Audience	Beginner Students
Due Date	To be confirmed by instructor
Format	Handwritten answers — scan and email your completed work

Instructions

Read each question carefully and research the answer before writing.

- All answers must be handwritten — typed submissions will not be accepted.
- Write in full sentences and use your own words.
- Include diagrams or examples where relevant — this will earn you bonus marks.
- Cite your sources at the bottom of each section.
- Scan your completed assignment and submit it before the due date.

Section A: Data Fundamentals

 **Research Tip:** Start with official documentation from platforms like IBM, Google Cloud, AWS, or Microsoft Azure — they offer excellent beginner-friendly explanations.

1. What is a Database? Define it in your own words and explain why databases are essential in the modern world. Provide two real-world examples of where databases are used.

A database is an organized collection of data, generally stored and accessed electronically from a computer system.



A spreadsheet system like Excel or Google sheets are the examples of database. Database tables are quite similar to spreadsheets. Each table has different columns which would contain different types of data.

2. What is a Data Warehouse? How does it differ from a regular database? Why would a company use a data warehouse? A data warehouse is a system designed to collect, store and analyze data from various business systems to provide a single source of truth for reporting and analytics.

Regular Database	Data Warehouse
- Daily transactions	- Reporting and analytics
- current operational data	- historical & consolidated data
- optimized for transactions	- periodic data loads.

3. What is a Data Lake? Explain the concept and describe how it differs from a data warehouse. When would a business prefer a data lake over a data warehouse?

A storage system that holds amounts of raw data in many formats structured (Tables, CSV), semi structured (JSON, XML, logs) and unstructured such as (images, videos, Audios, where as a data warehouse stores mostly structured, cleaned and organized data. A business would typically prefer a data lake over a data warehouse when it needs to store large volumes of diverse data and is not yet sure how that data will be used.

4. What is a Data Mart? How does it relate to a data warehouse and who typically uses it?

A data mart is a smaller, focused part of the warehouse build for one department or business area. Examples Sales mart, finance mart. Easier for business users, faster for reports, with department-specific KPIs and tighter security.

5. Explain the difference between Structured, Semi-Structured, and Unstructured Data. Provide one example of each.

Structured: A grid of data, like a spreadsheet, with a name. Each row is one record. Semi-structured: data does not fit neatly into rows and columns, but it still contains tags, labels example JSON file. Unstructured Data: Information does not follow a specific data model or structure and cannot be easily analyzed using traditional databases example PDF documents.



6. What is Data Modeling? Why is it important when designing a system that works with data? Describe

at least two types of data models. Data modeling is the process of defining, organizing and structuring data and the relationships between different data elements before a database or information system is built.

Data modeling creates a blueprint for how data will be stored, connected and used within a system. Data modeling is important because it helps organization, improve data quality, enhance system performance, reduce development costs and improve communication. Example: For a school finance system, data modeling

define a student her or his account. Example: For a school finance system, data modeling

7. What is Data Mining? How is it used in business? Provide two real-world examples of data mining in action. Data mining is the process of examining large amounts of data to discover hidden patterns, trends, relationships, and useful information that can help organization make better decisions.

Business use data mining to: improve customer understanding and increase sales and revenue. Real-world example: Data mining process, the system identifies patterns such as: customers who buy a laptop often buy a laptop bag and mouse.

8. What is a Database Management System (DBMS)? Name and briefly describe four popular DBMS platforms (e.g., MySQL, PostgreSQL, Oracle, MongoDB) and explain a suitable use case for each.

A Database Management system is a software that allows users to create, store, organize, retrieve, update and manage data efficiently.

Four popular DBMS Platforms: MySQL is an open-source relational database management system that stores data in tables consisting of rows and columns. Microsoft SQL Server is an enterprise-grade relational DBMS developed by Microsoft. Oracle Database is a powerful enterprise DBMS designed for large organizations requiring high availability and scalability. MongoDB is a NoSQL database that stores data in flexible document formats (JSON-like documents) rather than traditional tables.

9. What is ETL (Extract, Transform, Load)? Explain each step and describe why this process is important in data analytics.

ETL: Extract data from sources, transform it before loading, then load the cleaned result into the target warehouse. Classic approach used by traditional tools like Informatica and SSIS. Step 1: Extract: this involves collecting data from various sources. Example: Accounting system & student management system. Step 2: Transform: Phase cleans, validates and converts the data into a standard format. Example: Removing duplicate records, correcting errors.

Step 3: Load: Phase places the transformed data into the destination system. Example: Relates Agency reports or Revenue Dashboards.



10. What are common data storage formats used in analytics (e.g., CSV, JSON, Parquet, Avro)? Briefly describe each and state when you would use one over another.

1. CSV (Comma-separated values) A CSV file stores data as plain text, with each row representing a record and each column separated by commas. When to use: sharing data between systems, working with small to medium datasets.
2. JSON (JavaScript Object Notation) stores data in key-value pairs and supports nested structures. When to use: when working with web APIs, storing semi-structured data, exchanging data between applications.
3. XML (Extensible Markup Language) stores data using custom tags. When to use: integrating legacy systems, exchanging data between enterprise applications.
4. Parquet: is a columnar storage format designed specifically for big data analytics. When to use: working with large datasets, using data lakes.
5. Avro: is a row-based storage format that stores both data and schema together. When to use: streaming data between systems, processing real-time events.



Section B: Artificial Intelligence & Machine Learning

Research Tip: Explore resources like Google's Machine Learning Crash Course, IBM's AI explanations, or OpenAI's blog to build a strong foundational understanding.

11. What is Artificial Intelligence (AI)? Define it and explain the difference between Narrow AI (Weak AI) and General AI (Strong AI). Provide one example of each.

Artificial intelligence is a branch of a computer science that focuses on creating systems that can mimic human intelligence and perform tasks such as learning, reasoning, problem solving and decision-making.

Narrow AI (Weak AI) is designed to perform a specific task or a limited set of tasks. General AI (Strong AI) refers to a hypothetical AI system that can understand, learn, reason and apply knowledge across different tasks at a level comparable to a human being.

12. What is Machine Learning (ML)? How does it differ from traditional programming? Describe the three main types of Machine Learning: Supervised, Unsupervised, and Reinforcement Learning.

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed for every task. In traditional programming a programmer writes rules that tell the computer exactly what to do, whereas in machine learning, the computer learns the rules from historical data. 1. Supervised Learning: the model is trained using labeled data, meaning correct answers are already known. 2. Unsupervised Learning: data has no labels and the system tries to find hidden patterns.

13. What is Deep Learning? How does it relate to Machine Learning and Artificial Intelligence? What are Neural Networks and how do they work?

Deep learning is a subset of machine learning that uses multi-layered neural networks to automatically learn complex patterns and make predictions or decisions from large datasets. Artificial Intelligence (AI) - The broader field, goal, create machines that can mimic human intelligence, examples Expert systems. Machine Learning (ML) A subset of AI, goal, enable systems to learn from data without explicit programming. Neural Network is a computing model inspired by the structure of the human brain. A neural network is a collection of connected processing units that work together to learn relationships.

14. What is Natural Language Processing (NLP)? Describe what it is and provide three real-world applications of NLP that you use in everyday life.

NLP is the technology that allows computers to interact with human language by analyzing, understanding and responding to text or speech. Three real world applications: 1. Voice Assistants Examples: Siri, Google Assistant. When you say: 'Set an alarm for 8 a.m.' the assistant converts

the speech to text, understand your request, perform the action. 2. Email spam filters, Example: Gmail, Outlook, spam filters analyze incoming emails and determine whether they are important, promotional, spam.



15. What is a Large Language Model (LLM)? How is it different from a standard Machine Learning model? Name two well-known LLMs and explain what they are used for. A Large Language model (LLM) is an AI model that uses deep learning and neural networks to understand and generate human-like based on patterns learned from large datasets. GPT (Generative Pre-trained Transformer) is a family of large language models designed to generate human-like text and assist with a wide range of language tasks. Users: Chatbots, content creation, email drafting, coding assistance. Gemini is Google's family of large language capable of understanding and generating text, images and other forms of information.
16. What is Generative AI? How does it work at a high level? Name three tools that use Generative AI and describe what each one does. Generative AI is an AI technology that learns from large datasets and generates new content that resembles human-created content. At a high level, Generative AI works in four stages: Training, understanding the prompt, generating content and producing an output. Three popular Generative AI tools: 1. Microsoft Copilot, helps users work more efficiently by generating and assisting with Emails, Reports, PowerPoint presentations, Excel analysis. 2. ChatGPT is a conversational AI tool that generates human like text and assist with, answering questions, content creation, research assistance. 3. DALL-E generate images from text descriptions.
17. What is a Training Dataset? Why is the quality of training data so important when building an AI model? What could go wrong if the training data is biased or incorrect? A Training dataset is a set of examples that an AI model learns from in order to recognize patterns, make predictions, or generate outputs. There is a common saying in AI: 'Garbage in, Garbage out'. If the training data is poor quality, the AI model will produce poor results. High quality of data helps the model make correct predictions. Bias occurs when training data unfairly favours or disadvantages certain groups, outcomes or situations.
18. What is Computer Vision? How do machines 'see' and interpret images? Provide two real-world use cases of computer vision. Computer vision is an AI technology that allows machines to capture, process, analyze and interpret visual data from images and video to make decisions or perform tasks. Machine do not see images the same way humans do. Instead, the process images as collections of pixels. Step 1: Capture the image